# Inverter Chain

$$F = \frac{C_L}{C_{g_1}} : \quad C_L = \text{Load cap, } C_{g_1} = \text{Internal cap}$$

$$f = \sqrt[N]{F} : \quad f = \text{scaling factor, } N = \#\text{ inverters}$$

$$t_p = Nt_{p_0}\left(1 + \frac{f}{\gamma}\right) : \quad t_p = \text{prop delay}, t_{p_0} = \text{single prop delay}$$

$$\gamma = \frac{\text{intrinsic capacitance at output}}{\text{input gate capacitance}}$$

$$\sum C_L = \sum_{k=0}^{N}(f^N + \gamma f^N)C_{g_1} + C_L = C_{g_1}((1+\gamma)(1+f)+F)$$

Note: the formula above includes the input capcacitance.

$$P_{diss} = (\mathbf{freq})\alpha_{0\to1}V_{DD}^2\sum C_L : \quad \alpha_{0\to1} = \text{activity factor}$$

# Formula to Circuit

1. Get formula to the form $\bar{X}$
2. The pull-up of the circuit will be composed entirely of PMOS:
   - AND: parallel
   - OR: series

   The pull-down will be the inverse, that is NMOS:
   - AND: series
   - OR: parallel
3. To scale the transistors, the initial scale will be that of the minimized inverter. The multiplication factor of the transistors will be such that the resistance of each path in each netweork is the same as that of the corresponding path in an inverter.
   $C_{unit}$ is sum of the scaling of the PMOS and the NMOS for the given input.

# Gate Circuit

Let us assume that a fast inverter has a PMOS to NMOS width ratio of $a$. The following tells us what the electrical effort (**p**) and the logical effort (**g**) are:

| gate | $C_{unit}$ | p | g |
|---|---|---|---|
| inverter | $a+1$ | 1 | 1 |
| n-input NAND | $n+a$ | $n$ | $\frac{n+a}{a+1}$ |
| n-input NOR | $1+an$ | $n$ | $\frac{1+an}{a+1}$ |
| n-input MUX | $n(1+a)$ | $an$ | $a$ |
| n-input XOR | ? | $n*2^{n-1}$ | ? |

$$b = \frac{C_{on} + C_{off}}{C_{on}}$$

$$t_p = t_{p_0}\left(p + g\frac{f}{\gamma}\right)$$

$$B = \prod_i b_i \ ; \ G = \prod_i g_i$$

$$F = \frac{C_L}{C_{g_1}} : \quad C_L = \text{Load cap, } C_{g_1} = \text{Internal cap}$$

$$H = GFB$$

$$h = \sqrt[N]{H} = g_i f_i : \quad h = \text{gate effort, } f = \text{eff. fan-out}$$

$$S_n = \frac{hS_{n-1}}{b_{n-1}g_n} : \quad S_n = \text{scaling factor,}$$

$S_0 = 1$ unless otherwise stated.

$$D = t_{p_{tot}} = \sum_i t_{p_i} = t_{p_0}\left(\sum_{i=1}^{N} p_i \frac{Nh}{\gamma}\right)$$